

Topic 01: The Big Picture, Order, Linearity & Verifying Solutions

Professor Leonard Master Series · Lessons 1 to 4

Course: Applied Ordinary Differential Equations (ENGR 213 Companion)

1. Whiteboard Intuition: What is a Differential Equation?

In regular algebra, an equation asks:

"What number x satisfies this statement?"

In differential equations, an equation asks: **"What function $y(x)$ satisfies this relationship between itself and its derivatives?"**

Differential equations describe rates of change across nature:

- Fluid dynamics, heat transfer, and mechanical vibrations.
 - Electric circuit impedance and electromagnetic wave propagation.
 - Chemical reaction kinetics and population ecology.
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2. Core Definitions & Classifications

A. Ordinary vs. Partial Differential Equations

- **ODE (Ordinary Differential Equation):** Involves unknown functions of a **single independent variable** (e.g., dy/dx).
- **PDE (Partial Differential Equation):** Involves unknown functions of **multiple independent variables** (e.g., $\partial^2 u / \partial x^2 + \partial^2 u / \partial t^2 = 0$).

B. Order

The **order** of an ODE is the order of the **highest derivative** appearing in the equation.

- $y' + 3y = e^x \implies$ **1st order**.
- $y'' + 4y' + 9y = 0 \implies$ **2nd order**.
- $(y')^5 + y''' = x \implies$ **3rd order** (the exponent 5 affects degree, not order!).

C. Linearity

An n -th order ODE is **linear** if it can be written in the form:

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

Linearity Checklist:

1. The dependent variable y and all its derivatives $y', y'', \dots, y^{(n)}$ are of the **first degree** (power 1).
 2. The coefficients $a_i(x)$ and forcing function $g(x)$ depend **only on the independent variable x** (or are constants).
 3. No nonlinear functions of y exist (e.g., $\sin y, e^y, \ln y, \sqrt{y}$).
 4. No cross-products of y and its derivatives exist (e.g., $y \cdot y'$).
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3. Professor Leonard's Whiteboard Problem Walkthroughs

Problem 1.1: Verifying an Explicit Two-Parameter Solution

Statement: Verify that the two-parameter family $y(x) = c_1 e^{2x} + c_2 x e^{2x}$ is an explicit solution to the 2nd-order ODE:

$$y'' - 4y' + 4y = 0$$

Step-by-Step Whiteboard Solution:

- **Step 1: Compute the first derivative $y'(x)$ using the product rule:**

$$y'(x) = 2c_1 e^{2x} + c_2 (1 \cdot e^{2x} + x \cdot 2e^{2x}) = (2c_1 + c_2)e^{2x} + 2c_2 x e^{2x}$$

- **Step 2: Compute the second derivative $y''(x)$:**

$$y''(x) = 2(2c_1 + c_2)e^{2x} + 2c_2 (e^{2x} + 2xe^{2x}) = (4c_1 + 4c_2)e^{2x} + 4c_2 x e^{2x}$$

- **Step 3: Substitute y, y', y'' into the LHS of the ODE:**

$$\text{LHS} = y'' - 4y' + 4y$$

$$\text{LHS} = [(4c_1 + 4c_2)e^{2x} + 4c_2 x e^{2x}] - 4[(2c_1 + c_2)e^{2x} + 2c_2 x e^{2x}] + 4[c_1 e^{2x} + c_2 x e^{2x}]$$

- **Step 4: Group terms by e^{2x} and $x e^{2x}$:**
 - Coefficient of e^{2x} : $(4c_1 + 4c_2) - 4(2c_1 + c_2) + 4c_1 = 4c_1 + 4c_2 - 8c_1 - 4c_2 + 4c_1 = 0$.
 - Coefficient of $x e^{2x}$: $4c_2 - 8c_2 + 4c_2 = 0$.

$$\text{LHS} = 0 \cdot e^{2x} + 0 \cdot x e^{2x} = 0 = \text{RHS} \quad \checkmark$$

Conclusion: The solution is verified for all real values of c_1, c_2 .

Problem 1.2: Verifying an Implicit Relation and Checking Branches

Statement: Verify that the relation $x^2 + y^2 = 25$ is an implicit solution to $\frac{dy}{dx} = -\frac{x}{y}$ on the interval $-5 < x < 5$.

Step-by-Step Whiteboard Solution:

- **Step 1: Differentiate implicitly with respect to x :**

$$\frac{d}{dx}(x^2 + y^2) = \frac{d}{dx}(25)$$

$$2x + 2y \frac{dy}{dx} = 0$$

- **Step 2: Solve for dy/dx :**

$$2y \frac{dy}{dx} = -2x \implies \frac{dy}{dx} = -\frac{x}{y}$$

- **Step 3: Professor Leonard's Branch Analysis:** The relation defines two distinct explicit differentiable functions:

$$y_1(x) = \sqrt{25 - x^2}, \quad y_2(x) = -\sqrt{25 - x^2}$$

Both functions satisfy the differential equation on $(-5, 5)$. At $x = \pm 5$, $y = 0$, so dy/dx is undefined (vertical tangent lines).

Problem 1.3: Classification Drill (Order, Linearity, Homogeneity)

Statement: Classify each differential equation by order and linearity. If nonlinear, pinpoint the exact offending term:

1. $y''' + xy' + (\sin x)y = e^x$
2. $y'' + yy' = 2x$
3. $\frac{d^2 y}{dx^2} + \sqrt{1 + \left(\frac{dy}{dx}\right)^2} = 0$
4. $\frac{d^4 y}{dx^4} + x^2 y = \ln x$
5. $y' + \cos(y) = 0$

Step-by-Step Whiteboard Solution:

1. **3rd Order, Linear:** Highest derivative is y''' . The coefficients 1 , x , $\sin x$ depend only on x . y and its derivatives are to power 1.
2. **2nd Order, Non-linear:** Highest derivative is y'' . The term yy' is a product of the dependent variable and its derivative!
3. **2nd Order, Non-linear:** Highest derivative is y'' . The derivative dy/dx is squared inside a square root: $\sqrt{1 + (y')^2}$.
4. **4th Order, Linear:** Highest derivative is $y^{(4)}$. The coefficient x^2 depends only on x , and y is linear.
5. **1st Order, Non-linear:** The dependent variable is inside a trigonometric function: $\cos(y)$.

Problem 1.4: Initial Value Problem (Finding the Specific Curve)

Statement: Given the general solution $y(x) = Ce^{-3x} + \frac{1}{3}x - \frac{1}{9}$ to $y' + 3y = x$, find the unique particular solution satisfying $y(0) = 2$.

Step-by-Step Whiteboard Solution:

- **Step 1: Set $x = 0$ and $y = 2$ in the general solution:**

$$2 = Ce^{-3(0)} + \frac{1}{3}(0) - \frac{1}{9}$$
$$2 = C(1) + 0 - \frac{1}{9} \implies C = 2 + \frac{1}{9} = \frac{19}{9}$$

- **Step 2: Write the unique particular solution:**

$$y(x) = \frac{19}{9}e^{-3x} + \frac{1}{3}x - \frac{1}{9}$$

Problem 1.5: Piecewise Differentiability at Junctions (Leonard Conceptual Master)

Statement: Consider the function $y(x) = \begin{cases} x^4, & x \geq 0 \\ -x^4, & x < 0 \end{cases}$. Does this function satisfy $xy' - 4y = 0$ for all real $x \in (-\infty, \infty)$?

Step-by-Step Whiteboard Solution:

- **Step 1: Differentiate for $x > 0$:** $y' = 4x^3$. Substituting into ODE: $x(4x^3) - 4(x^4) = 4x^4 - 4x^4 = 0$. Valid!
- **Step 2: Differentiate for $x < 0$:** $y' = -4x^3$. Substituting into ODE: $x(-4x^3) - 4(-x^4) = -4x^4 + 4x^4 = 0$. Valid!
- **Step 3: Check differentiability at $x = 0$ using limits of difference quotients:**

$$y'(0) = \lim_{h \rightarrow 0} \frac{y(h) - y(0)}{h}$$

- Right limit: $\lim_{h \rightarrow 0^+} \frac{h^4 - 0}{h} = \lim_{h \rightarrow 0^+} h^3 = 0$.
- Left limit: $\lim_{h \rightarrow 0^-} \frac{-h^4 - 0}{h} = \lim_{h \rightarrow 0^-} -h^3 = 0$. Since both one-sided derivatives match, $y'(0) = 0$.
- **Step 4: Check ODE at $x = 0$:** $0 \cdot y'(0) - 4y(0) = 0(0) - 4(0) = 0$. **Conclusion:** Yes! $y(x)$ is a valid solution over the entire real line $(-\infty, \infty)$.

4. Common Exam Traps & Professor Leonard Warnings

- **Trap 1:** Calling $(y')^4$ a 4th-order ODE. It is 1st-order, 4th-degree! Order is derivative count, not power.
- **Trap 2:** Missing non-linearity when x and y switch roles. dx/dy might be linear in $x(y)$ even when dy/dx is non-linear in $y(x)$!
- **Trap 3:** Forgetting that general solutions to an n -th order linear ODE must possess exactly n essential arbitrary constants.